WINDOWS HDD FILE RECOVERY SUITE

USER MANUAL | VERSION 2.0

A practical guide to connecting the Windows Recovery Agent, scanning for deleted files, and restoring them safely.

RECOVER DELETED FILES WITH CARE

This manual covers the current hosted control panel and the local Windows Recovery Agent. The website can be hosted on Vercel; the agent must run on the same Windows PC that opens the site, because it reads that PC's disks.

Before you scan
Stop using the affected drive. Prepare a separate physical drive for recovered files. Read the safety rules in this guide before starting.

Prepared for the Windows HDD File Recovery Suite project

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At a glance

Use this guide in order the first time. The web interface is a control panel; low-level disk access is provided by a separate agent running locally on Windows.

Five-minute setup checklist

1. **Open the hosted site on the Windows PC**
   The agent address 127.0.0.1 refers to the computer using the browser. Opening the site on a phone or another computer will not reach an agent on your Windows PC.

2. **Download and extract the agent ZIP**
   Use Windows Setup & ZIP. Extract the package to a normal folder such as C:\RecoveryAgent.

3. **Run the launcher as administrator**
   Right-click start-agent.bat, choose Run as administrator, and approve the Windows prompt. Keep its terminal window open.

4. **Allow the browser to reach localhost**
   If the browser asks for local network access, allow it for the site. Then open Live Setup Verification and choose Test Connection Now.

5. **Check all three readiness signals**
   Agent API should be Online, C++ Engine should be Ready, and Administrator Rights should be Elevated. A Connected label alone only confirms the API responded.

Understand the status labels
MOCK DATA means the interface is showing built-in sample content. Connected means the local API answered. For physical disk recovery, also confirm Engine is Ready and Administrator Rights is Elevated.

In this manual

Safety and system model | First-time agent setup | Account access | Scan workflow | Results and restore | History and reports | Troubleshooting | Limits and accessibility

Safety first

Deleted data may remain on a drive until other data overwrites it. The safest first action is to prevent further writes to that drive. Recovery is never guaranteed, and a scan can take hours.

Before you scan

- **Stop using the affected drive.** Do not download, install, copy, or save files to it. Avoid routine use of the Windows system drive if it is the source.
- **Do not format, initialize, repair, or run CHKDSK.** These actions may change metadata needed for recovery.
- **Choose a physically separate destination disk.** A second partition on the same physical disk is not safe. Do not restore files to the disk being scanned.
- **For a clicking, overheating, disconnecting, or rapidly failing disk, stop scanning.** Repeated reads can worsen the situation. Consider professional recovery or a sector image.
- **Do not treat confidence scores as guarantees.** Preview important files and verify restored copies.

> **Read-only source access**
> The engine is designed to open source disks for reading and to write recovered copies only to the chosen destination. The destination safety check is a safeguard, not a substitute for selecting a different physical disk.

Why restore to another physical disk?

If a recovered file is written onto free space on the source disk, that write can overwrite other deleted files that have not yet been recovered. A separate USB or internal disk avoids this conflict. Confirm the destination has enough free space before restoring.

How the hosted site and local agent work

The Vercel website does not have direct access to the disks attached to your PC. The local Windows Recovery Agent provides that access. Your browser connects to the agent at http://127.0.0.1:8000 or http://localhost:8000; this traffic stays on the Windows PC.

Part	What it does
Hosted website	Displays the recovery wizard, session history, results, and setup instructions.
Browser on Windows PC	Connects to the site and separately to the local agent. It must be on the same PC as that agent.
Recovery Agent	Local API service that reports its status and coordinates scans.
Native C++ Engine	Performs disk-level scanning when available and ready.
Destination disk	Receives recovered file copies after you choose a restore location.

Three checks are different
A successful Test Connection is not proof of live disk scanning. Verify Agent API = Online, C++ Engine = Ready (not Simulated), and Administrator Rights = Elevated. Then check the site-wide mode indicator: it should no longer say MOCK DATA.

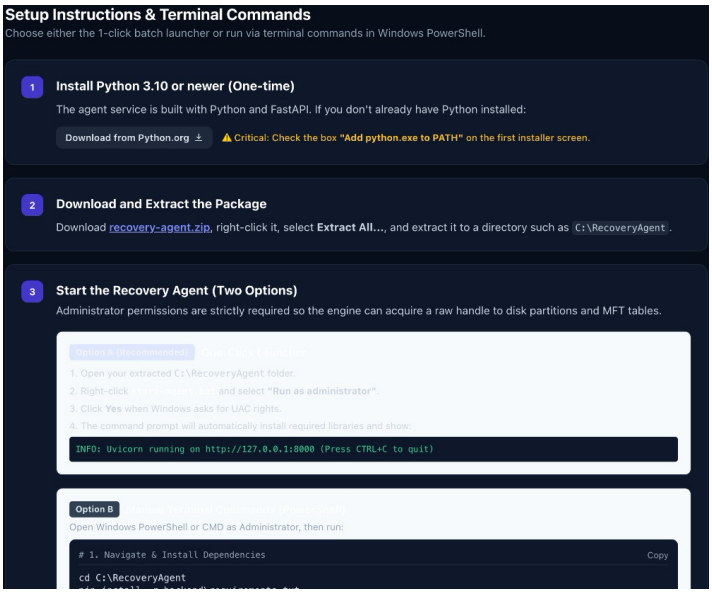
Mode and status guide

Indicator	Meaning	What to do
MOCK DATA	The web app is using sample/demo data; no real disks are being read.	Fix the app configuration or live agent mode before scanning.
Connected / API Online	The browser reached the local agent API.	Also verify engine readiness and elevated rights.
Engine Simulated	The API is responding but the native scan engine is not active.	Check the downloaded package and agent terminal.
Administrator Rights: Standard	The service is not elevated. Raw disk access may fail.	Stop the service and restart the BAT launcher as administrator.
Ready + Elevated + live mode	The key prerequisites for physical disk scans are present.	Continue to the Recover page and confirm actual drives appear.

Set up the Windows Recovery Agent

Do this on the Windows 10 or 11 (64-bit) PC whose drives you want to scan. The current download is a ZIP package containing the local bridge, native engine, launcher, and database files.

1	**Download the package** Sign in to the hosted site, open Windows Setup & ZIP, and download recovery-agent.zip. Use the official project site or the download link presented inside it.
2	**Extract the ZIP** In File Explorer, right-click the ZIP and select Extract All. Extract it to a simple local folder, for example C:\RecoveryAgent. Do not run the launcher from inside the ZIP preview.
3	**Start with administrator rights** Open the extracted folder, right-click start-agent.bat, and choose Run as administrator. Approve the User Account Control prompt. The account being an administrator is not enough by itself; the running process must be elevated.
4	**Keep the service running** Leave the command window open. It should report that the service is running at http://127.0.0.1:8000. Closing the window stops the agent.
5	**Verify in the browser** Return to Windows Setup & ZIP, allow local network access if asked, and click Test Connection Now. Check API, Engine, and Administrator Rights separately.

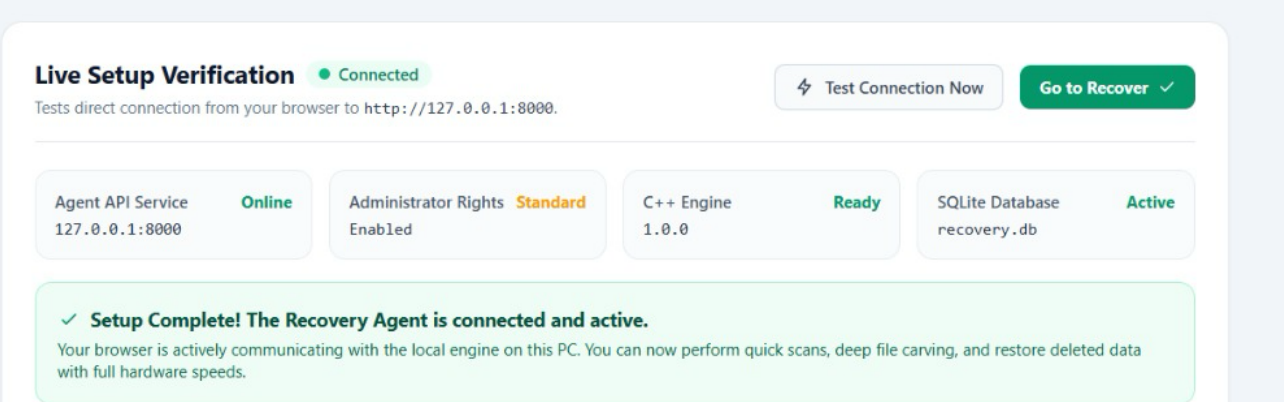


Setup instructions shown in the application. The ZIP package is the simplest path; the terminal instructions are an alternative for advanced users.

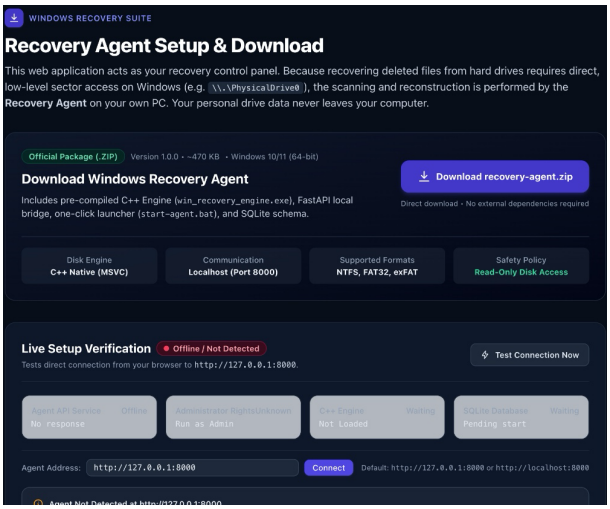
Confirm the agent is ready

A local service can be reachable while still unable to perform physical disk recovery. Use the diagnostic panel as a checklist, not just the green Connected badge.

Check	Expected for live disk scan	If it differs
Agent API Service	Online	Keep the terminal open; verify address and browser local-network permission.
Administrator Rights	Elevated	Stop the process and start start-agent.bat with Run as administrator.
C++ Engine	Ready	If Simulated or Waiting, confirm the extracted package includes the native engine and restart the agent.
SQLite Database	Active / Ready	Restart agent; review startup messages for database errors.
Site-wide mode	Live mode, no MOCK DATA badge	If MOCK DATA remains, the app is configured to use mock data or its live-mode detection has not updated. Refresh after the agent is ready.



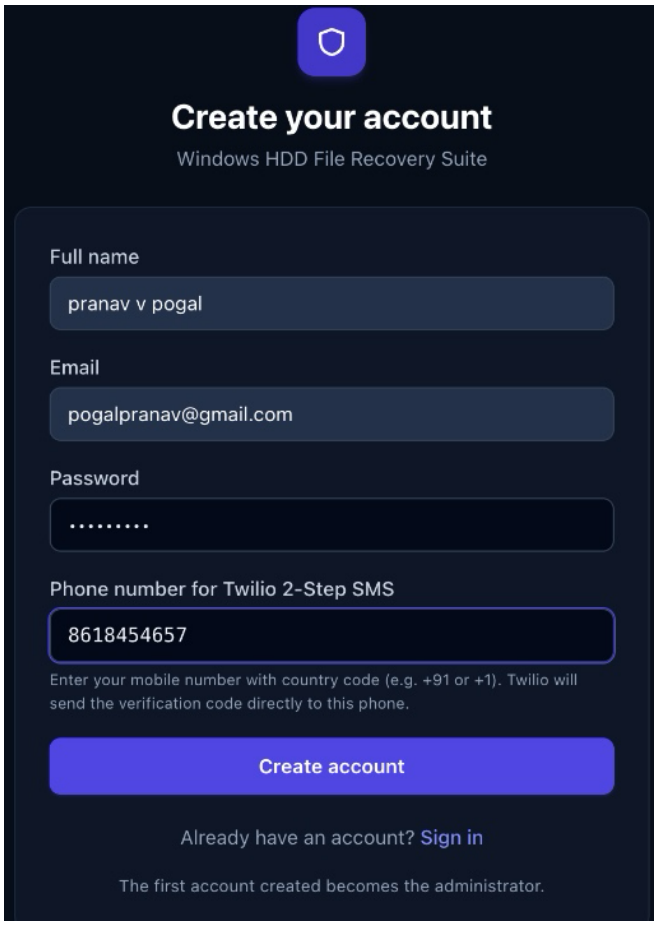
Example status panel: the API and engine are available, but this example shows Administrator Rights as Standard. That is not elevated and can block physical disk access.



Example of an offline setup panel. Use Test Connection Now after starting the agent and granting browser access.

Create an account

Open the account form and enter a name, email address, password, and phone number with country code. Complete the requested fields and choose Create account. Follow any confirmation shown by the application.



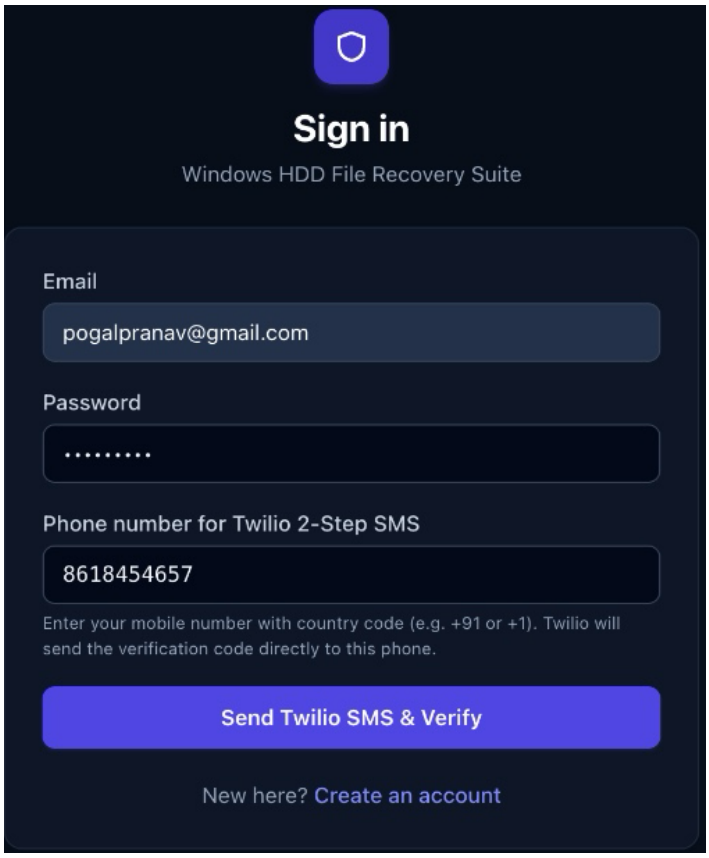
Example account-creation form. The names and contact details shown in this supplied screenshot are fictitious, as confirmed by the project owner.

Keep account access secure

Use a password you do not reuse elsewhere. Keep the SMS phone number accessible for sign-in, and do not share a verification code. Account creation and agent setup are separate: creating an account does not start or elevate the local Recovery Agent.

Sign in with SMS verification

The sign-in page asks for your email, password, and phone number. Use the phone number with its country code so the verification message can be delivered.



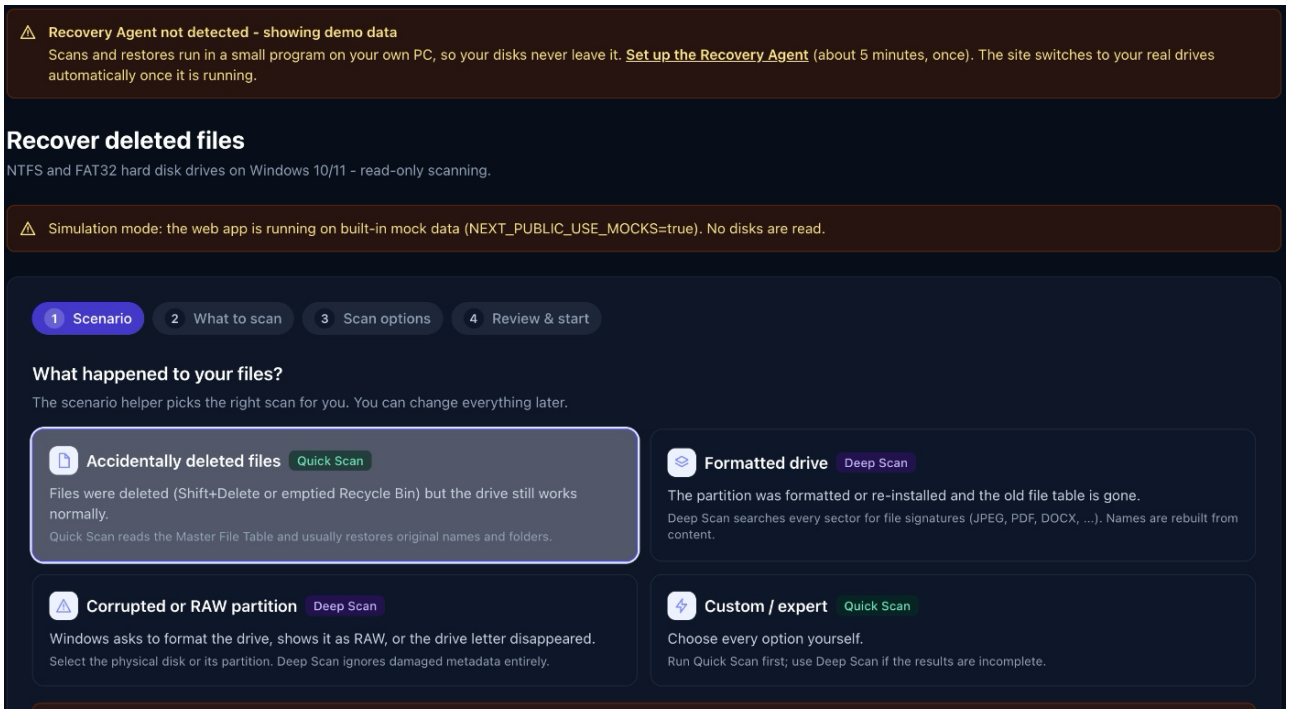
Example sign-in form with SMS verification. Values in this supplied screenshot are fictitious, as confirmed by the project owner.

1	**Request the code** Choose Send Twilio SMS & Verify and wait for the six-digit SMS code.
2	**Enter the code** Type the code into the verification field and submit it before it expires.
3	**If the SMS does not arrive** Check the country code and phone number, wait briefly, then use the resend option if available.

Agent status is separate
Successful sign-in does not switch the site from MOCK DATA or prove the agent is ready. Verify LIVE AGENT, Engine Ready, and Administrator Rights Elevated on the setup and recovery screens.

Choose the right scan

Open Recover and choose the scenario closest to what happened. The scenario helper recommends a starting point; you can change the scan scope and options before starting.



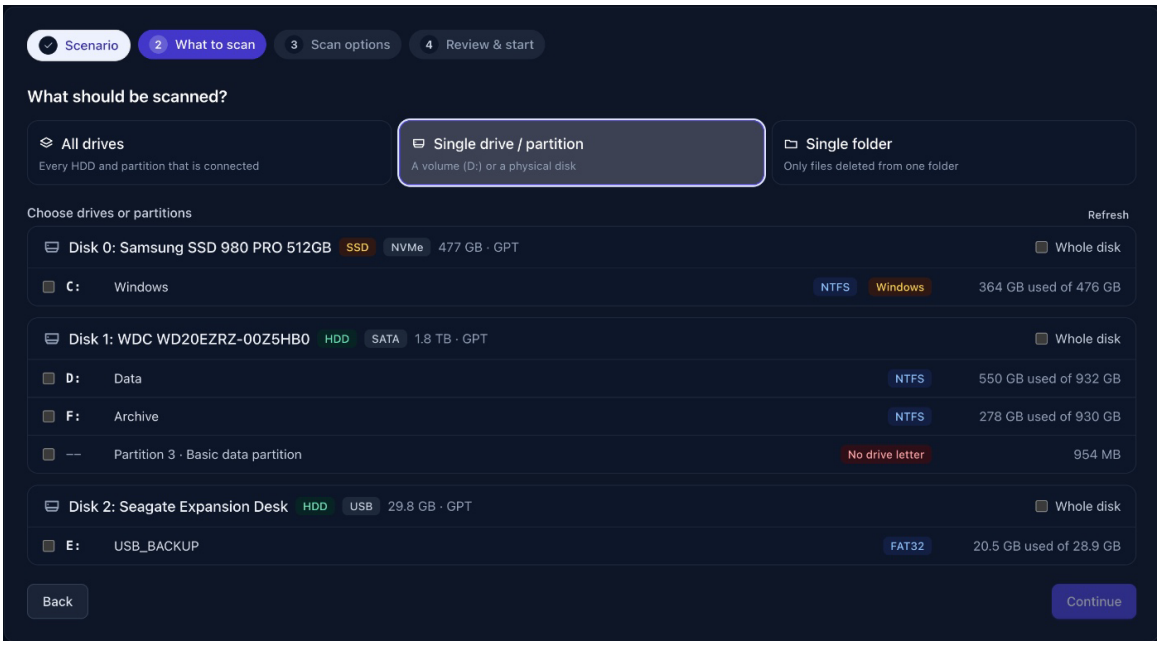
Example recovery scenario chooser. This supplied screenshot shows MOCK DATA and is for interface guidance only; it does not show actual attached drives.

Situation	Start with	Notes
Recently deleted files; drive otherwise works	Quick Scan	Reads file-system metadata and often retains names and folders.
Drive formatted or file table damaged	Deep Scan	Looks for known file signatures across sectors; may take hours.
RAW partition or missing drive letter	Select the affected partition or physical disk; try Deep Scan	Choose carefully; avoid formatting prompts.
One deleted folder	Single folder, usually Quick Scan	Enter the former folder path if the interface requests it.
Unsure	Quick Scan, then Deep Scan if needed	Deep scanning can take much longer and may return reconstructed names.

Select scan scope and options

Choose what to scan

All drives covers all detected eligible drives. **Single drive / partition** lets you select a volume or physical disk. **Single folder** narrows a scan to one folder path. Use Refresh if you have connected or disconnected a drive since opening the page.

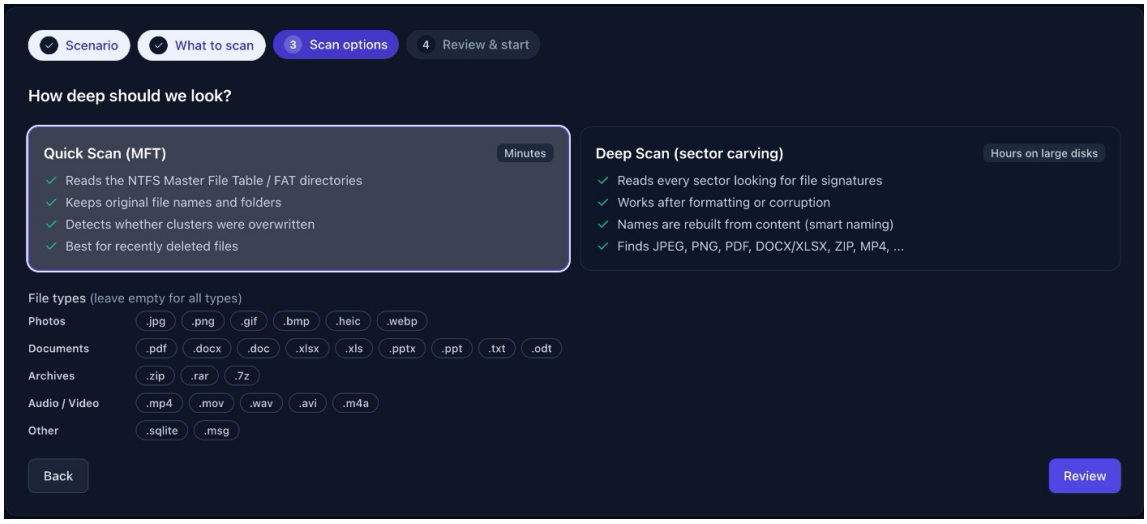


Illustrative drive-selection screen supplied in MOCK DATA mode. Drive names and capacities shown here are sample values, not a report of your PC.

Choose scan depth and file types

Quick Scan (MFT / FAT metadata)	Deep Scan (sector carving)
Usually completes in minutes on supported volumes. Reads file-system records and can preserve original names and folders.	May take hours on large disks. Searches sectors for file signatures and may rebuild names from content.
Best starting point for recent deletions when metadata remains readable.	Useful after formatting, corruption, or incomplete Quick Scan results. Fragmented files may be incomplete.

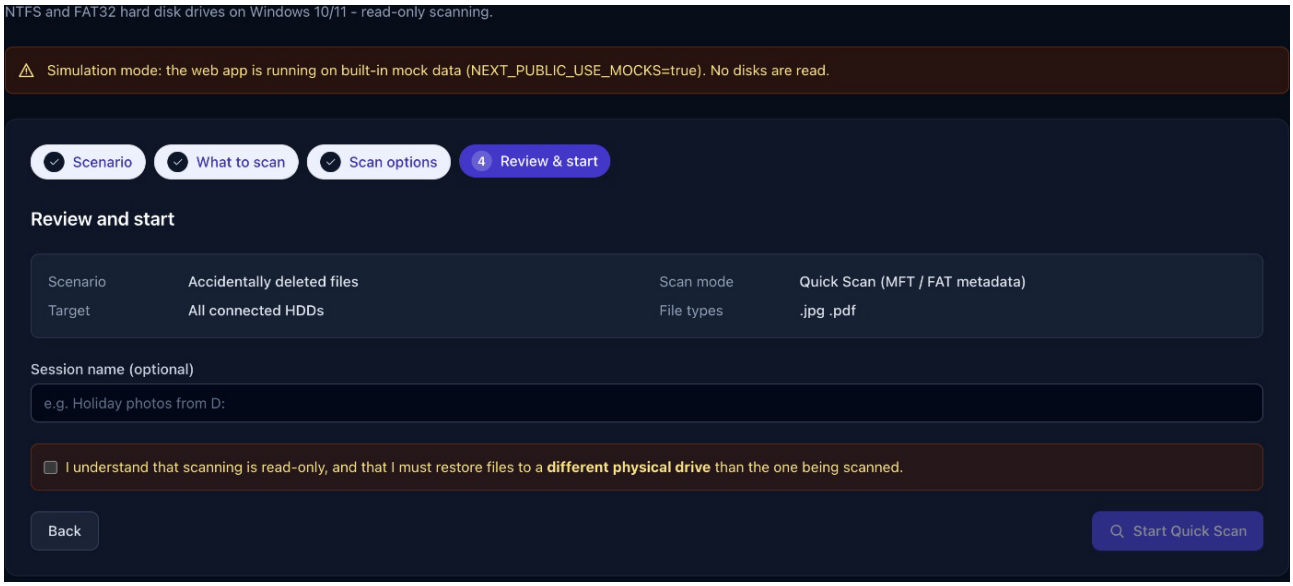
File-type filters are optional. Leave them empty to include all types. Otherwise select the extensions you need, such as .jpg, .pdf, or .docx. Filters limit returned results; they do not improve the condition of overwritten data.



Example scan options and extension filters. The screen is from mock mode and only illustrates the controls.

Review and start the scan

Before starting, check the scenario, selected source, scan mode, and file-type filters. If anything is wrong, go back and change it. Give the session a short name if you want to find it later.



Review screen example. It shows the mock-data warning and a safety acknowledgement before scanning.

Safety acknowledgement

Read the confirmation carefully. It asks you to acknowledge that scanning is read-only and that restored files must go to a different physical drive. Check it only after you have prepared the destination.

While scanning

- Keep the browser tab and Recovery Agent running. Do not disconnect the source or destination disks.
- Watch progress and read-error messages. A failing drive should not be subjected to repeated long scans.
- A Deep Scan can take hours. Avoid sleep, restart, or power loss during a scan.
- If you stop early, keep the session if you want to review partial results; check the interface for its current save behavior.

Review scan results

When the scan completes, narrow results using search, file type, path, or the available filters. Open Preview when available to inspect a file before restoring it. Deep Scan results may have reconstructed names because the original file-system record was missing.

Understand confidence

Rating	What it suggests	How to use it
High	The file signature and available content checks look promising.	Preview and restore important files first.
Medium	The file may be partial, fragmented, overwritten in places, or not fully verified.	Restore to the separate destination and open the copy to check it.
Low	The content may be damaged or incomplete.	Treat as a last attempt; do not assume it will open.

A score is an estimate
A confidence rating is a guide based on checks the software can perform. It cannot prove that every byte is correct or that a file will open in its original application.

If a file is missing

Try a Deep Scan if you ran Quick Scan and the file-system metadata may have been lost. If a Deep Scan does not find it, the data may have been overwritten, trimmed by an SSD, encrypted, fragmented, or in a format the engine does not recognize.

Restore files safely

1. **Select the files**
   Select one or more candidates. Check names, paths, sizes, previews, and confidence ratings where available.

2. **Choose the destination**
   Choose a folder on a physically different disk from the source. Confirm free space. Do not choose another partition on the same physical disk.

3. **Start restore and wait**
   Keep the destination connected. The source is read and copies are written to the destination folder.

4. **Verify the restored copies**
   Open key files from the destination. Check documents, photos, and archives in their normal applications. Save an additional backup if the files matter.

Do not restore to the source
Writing recovered data onto the scanned physical disk can overwrite other deleted data. A different drive letter is not enough if both letters belong to partitions on the same physical disk.

Name conflicts and folders

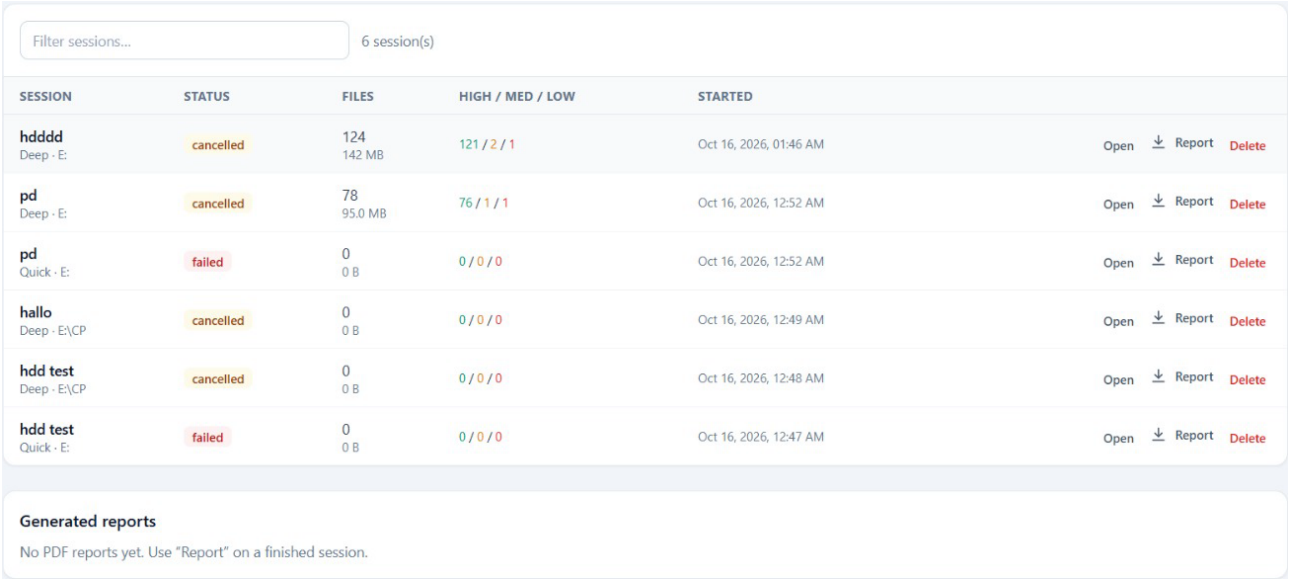
The application may add a safe suffix where a file with the same name already exists, and may offer to recreate folder paths. Review the restore summary and destination carefully; preserve existing files at the destination.

History, search, and reports

Use History & Reports to find earlier scan sessions, review their summaries, and download available reports. Filter sessions by name or date if the list is long.

Session list

Each row summarizes a session, including its status, file count, scan mode, target, and start time. Choose Open to review a session, Report to download its report, or Delete to remove its history entry. A cancelled or failed session may have no recoverable results.



Example History & Reports page in LIVE AGENT mode. Session names, counts, dates, and statuses shown are fictitious demonstration data.

Search, import, and reports

Use the session filter to find a saved scan. If your workflow provides a candidates.json file, import it through the control shown in History & Reports; imported candidates are metadata, not proof that file bytes remain intact. Reports can contain disk labels and file names, so store and share them carefully.

Deleting a session
Check the on-screen confirmation before deleting history. A session record is distinct from a restored file copy; deleting history should not be used as a way to erase or secure restored data.

Troubleshooting

Symptom	Checks
Site still says MOCK DATA	The app is still running in mock mode. Agent connection alone does not switch that banner. Confirm deployment configuration and live-mode detection, then reload the page after the agent is ready.
Test Connection says Connected, but drives are sample entries	Check C++ Engine = Ready, Admin Rights = Elevated, and the site-wide mode is live. On Recover, refresh the drive list. Sample drives are not physical disks.
API Online, Admin Rights = Standard	Stop the current agent. Right-click start-agent.bat and choose Run as administrator; accept UAC. Retest the panel.
Browser cannot reach 127.0.0.1:8000	Confirm the agent terminal is still open and listening. Allow local network access for the hosted site. Check the entered address and try localhost:8000 if needed.
Agent works locally but not from hosted site	Open the Vercel site in a browser on the same Windows PC running the agent. localhost on another device points to that other device. Allow local network access for the site.
Engine shows Simulated or Waiting	Confirm the ZIP was fully extracted and includes the engine file. Restart the agent and inspect startup errors.
Access denied / administrator required	Restart the service elevated. Close older non-elevated agent windows so the browser does not reconnect to the wrong instance.
Quick Scan finds nothing	Try Deep Scan if appropriate. Avoid writing to the source and check the correct volume or physical disk was selected.
Deep Scan is slow or has read errors	Large disks take hours. For repeated errors, unusual noises, or disconnects, stop and seek professional help.
SMS code does not arrive	Check the country code and phone number, wait briefly, and use resend if offered. Never share the code.

If the problem persists, record the exact status labels and error message. Remove personal information, phone numbers, passwords, verification codes, and sensitive drive names before sharing screenshots.

Limits, accessibility, and references

Recovery limits

Recovery depends on whether the deleted data has been overwritten and whether the file system, device, and file type are supported. SSD TRIM can make deleted data unavailable quickly. Encryption, fragmentation, failing hardware, and unsupported formats can reduce results. No scan can guarantee recovery.

The application advertises NTFS, FAT32, and exFAT support. Quick Scan uses file-system metadata; Deep Scan searches for supported signatures. Use the status panel and current application screens for the installed engine version and available capabilities.

Accessibility and document design

This manual uses a clear heading hierarchy, readable body type, strong text contrast, descriptive captions, and text explanations alongside screenshots. These practices are informed by the Government of India MeitY Digital Brand Identity Manual (v3, January 2025) and GIGW 3.0 accessibility principles. This project is not an official government service, and this document does not claim formal GIGW certification.

The MeitY guidance recommends left-aligned body text for government digital content. The body copy in this manual is fully justified to follow the project owner's explicit layout request; headings, tables, lists, and captions remain left aligned for scanning and accessibility.

Glossary

Term	Meaning
Agent	The local Windows service that connects the browser to disk recovery functions.
MFT	Master File Table used by NTFS to record files and their locations.
Carving	Finding files by recognizing byte patterns rather than relying on file-system records.
Physical disk	A complete storage device; multiple drive letters or partitions can exist on one physical disk.
Mock data	Built-in sample content for demonstration; it does not represent real disks or scan results.
Elevated	A process running with Windows administrator privileges after UAC approval.

Reference materials: previous suite manual; project UI and current setup/recovery screens; Government of India MeitY Digital Brand Identity Manual, version 3 (January 2025); GIGW 3.0 accessibility principles.

Quick reference

Before scan
Stop using source disk. Confirm LIVE mode, API Online, Engine Ready, and Admin Rights Elevated. Select the correct source and a different physical destination.

Do	Avoid
Act quickly and minimize writes to the source.	Installing apps or saving downloads to the affected disk.
Start with Quick Scan for recent deletion.	Assuming a green Connected label proves live scan readiness.
Use Deep Scan when metadata is damaged or results are incomplete.	Formatting or repairing a drive before recovery.
Restore to a different physical disk and verify the files.	Restoring to another partition on the same physical disk.
Treat demo screenshots and confidence scores as examples and estimates.	Treating sample drive entries as the drives attached to your PC.

Windows HDD File Recovery Suite

For the latest setup details, open Windows Setup & ZIP in the application and review the live status panel. Stop if the disk is physically failing or the data is irreplaceable.